AURA

AI-Powered Jewellery Visual Search Engine

Global Institute of Science & Technology

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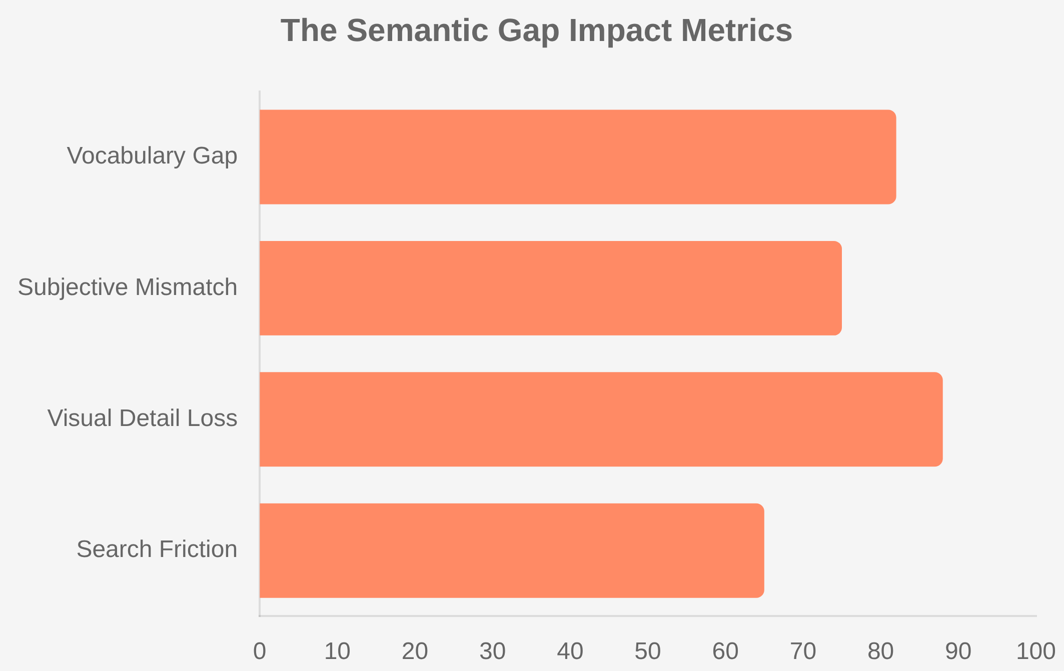
The Semantic Gap in Discovery



Source: E-commerce Visual Search Trends Report 2024

Vocabulary Mismatch

Users lack professional terminology (e.g., "filigree", "pavé") used in database tags, creating a significant discovery gap.

Subjectivity Bias

Terms like "elegant" or "vintage" are interpreted differently by users and systems, resulting in inconsistent outcomes.

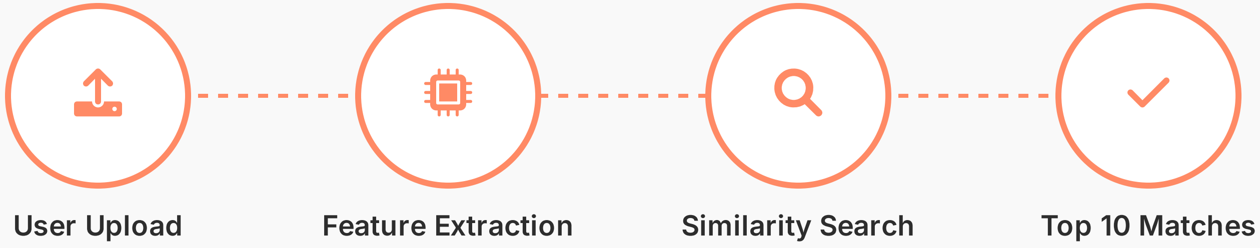
Visual Complexity

Intricate details and textures of jewellery are nearly impossible to translate into text strings accurately.

Search Inefficiency

Traditional keyword search leads to "zero results" for specific visual intents that are hard to describe.

Introducing AURA



"See it. Search it. Find it."

Direct Upload

Upload any photo instead of typing descriptive keywords.

AI Analysis

Deep learning extracts 1,280 visual feature points.

Vector Search

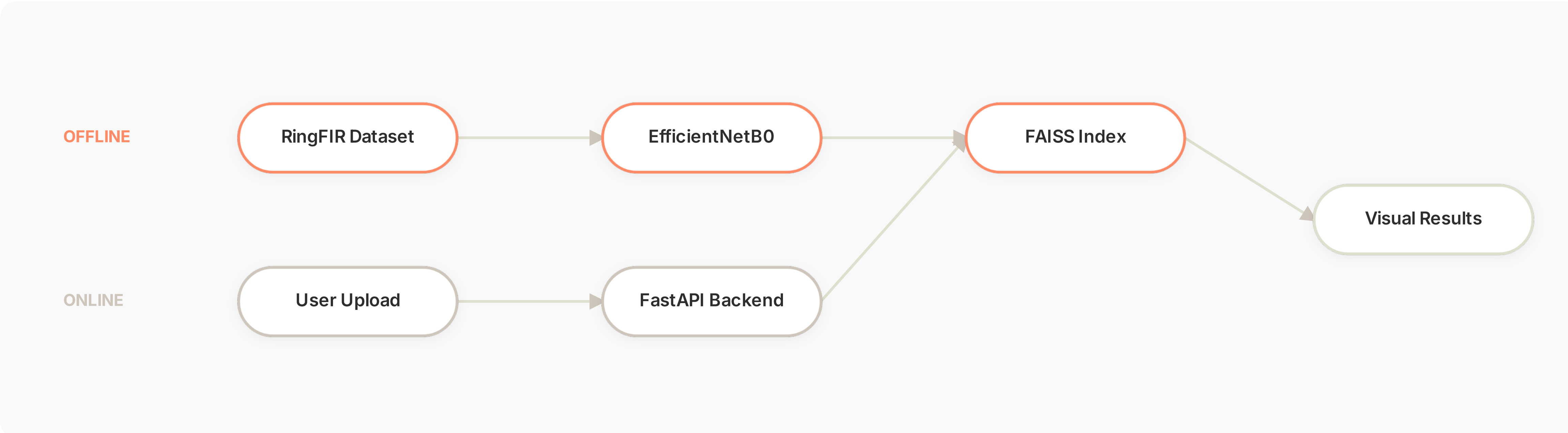
FAISS identifies nearest neighbors in under 50ms.

Core Value Proposition

AURA eliminates the "semantic gap" by matching visual intent directly to visual assets, bypassing the need for manual tagging or text descriptions.

System Architecture

Dual-Phase Processing: From Offline Indexing to Real-Time Discovery



Offline Phase

RingFIR dataset ingestion. EfficientNetB0 extracts 1,280-dim vectors from 2,605 images, compiled into a FAISS IndexFlatL2.

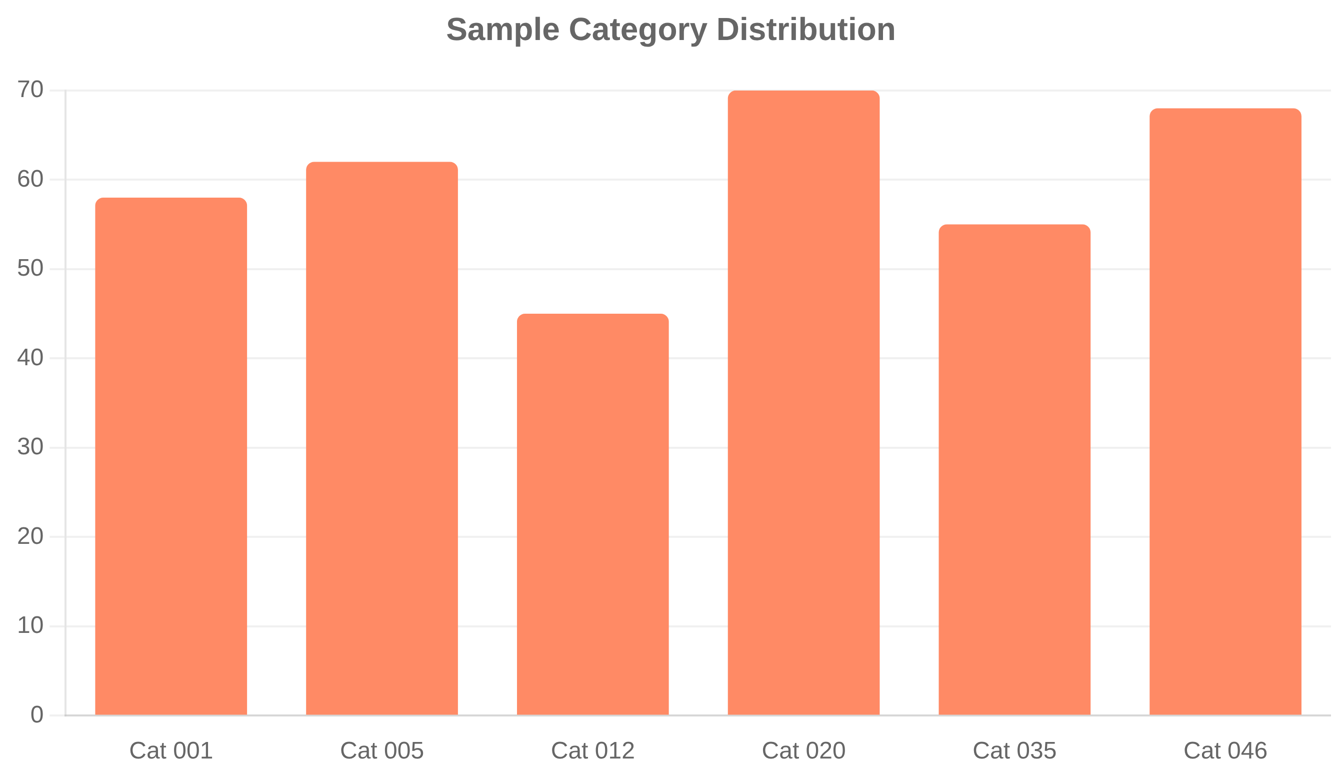
Online Phase

FastAPI backend receives query uploads. Real-time extraction and FAISS similarity search identify neighbors in <50ms.

Integration

Frontend renders result cards via Vanilla JS. Glassmorphism UI provides a premium, responsive discovery experience.

Dataset — RingFIR



Source: RingFIR Jewellery Image Dataset (Earring Collection)

2,605

Total Earring Photographs

46

Distinct Design Categories

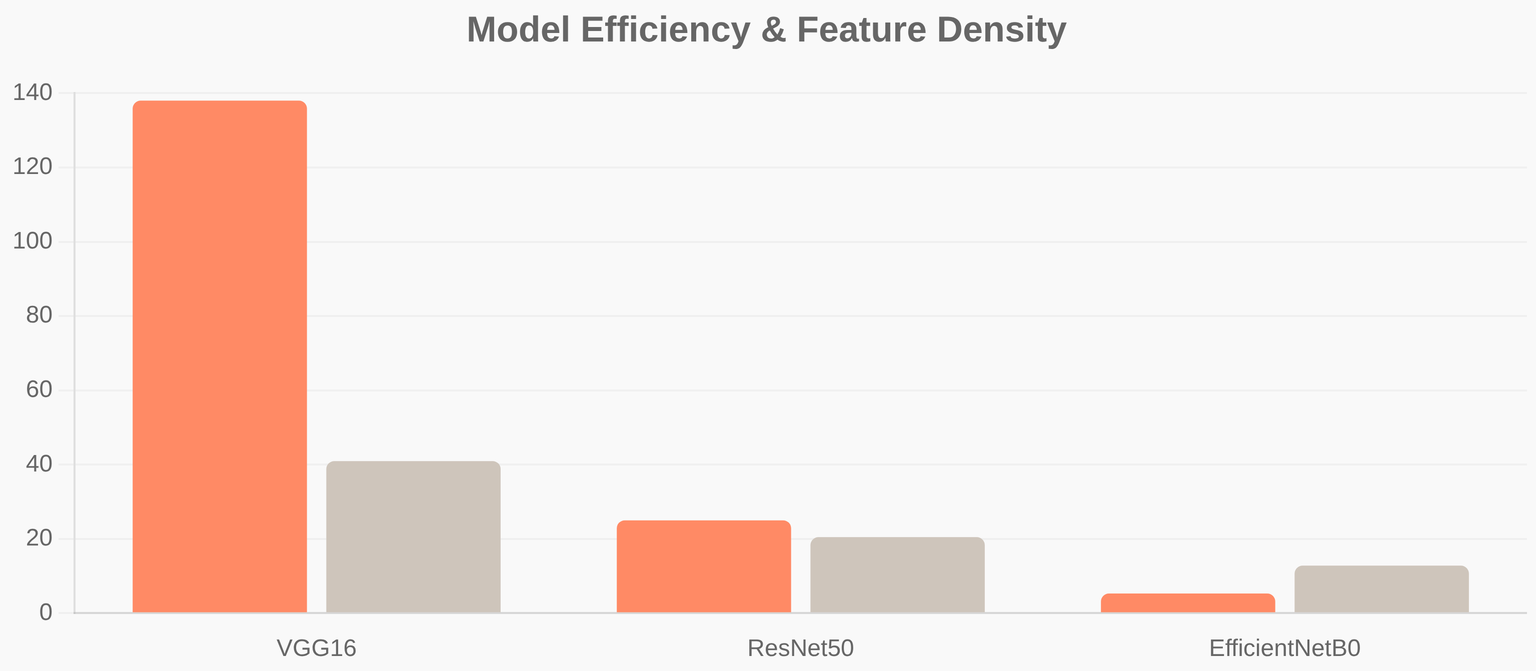
224 × 224

Standardized Input Resolution

100%

Automated Pipeline (No Manual Tagging)

The Visual Fingerprint



Source: Google Brain Research - EfficientNet Architecture

Feature Extraction:

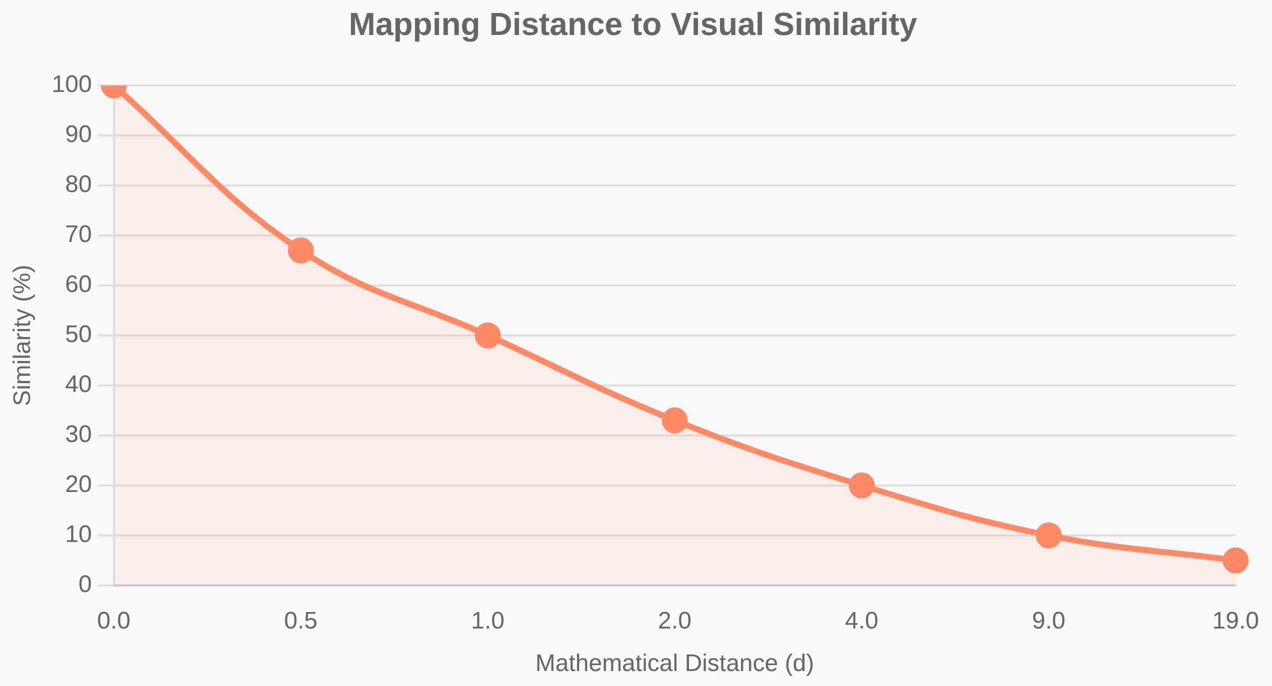
Efficiency (Parameters)

Density (1280 Numbers)

The AI "Eye"

- **Feature Extraction**: Translates pixels into a list of 1,280 descriptive numbers.
- **Visual Essence**: Captures curvature, texture, and gem settings.
- **Transfer Learning**: Uses Global Average Pooling to yield compact embeddings.

The Digital Warehouse



FAISS builds an optimized index of the 2,605-vector map.

2,605

Vectors plotted in the multi-dimensional map

IndexFlatL2

Exhaustive search for perfect accuracy

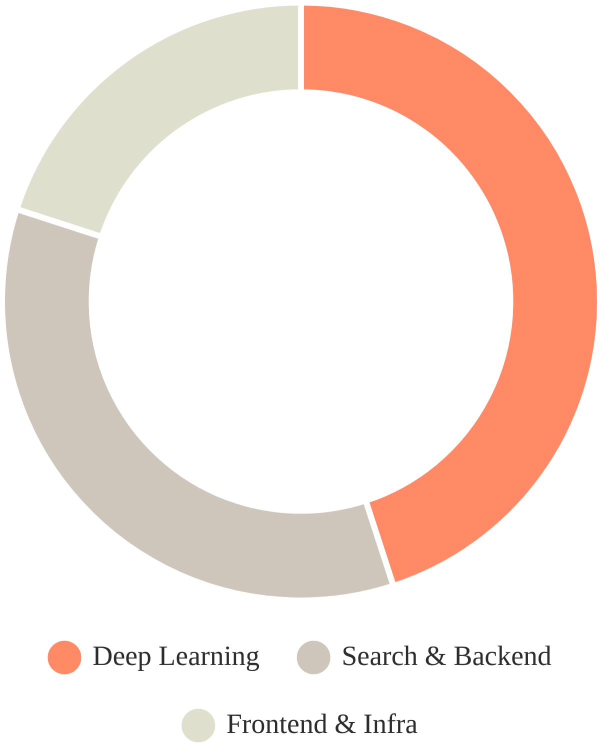
The "Filing Cabinet" Logic

$$S = (1 / (1 + d)) \times 100\%$$

FAISS measures mathematical distance between fingerprints. Closest neighbors are identified as the perfect visual matches.

FAISS (Facebook AI Similarity Search) provides a super-fast digital filing cabinet for near-instant retrieval.

Technology Stack



Deep Learning

CORE INTELLIGENCE

TensorFlow & Keras for loading the pre-trained **EfficientNetB0** model to perform high-dimensional feature extraction.

Search & Backend

PERFORMANCE LAYER

FastAPI for async REST services and **FAISS (Meta AI)** for sub-millisecond vector similarity search.

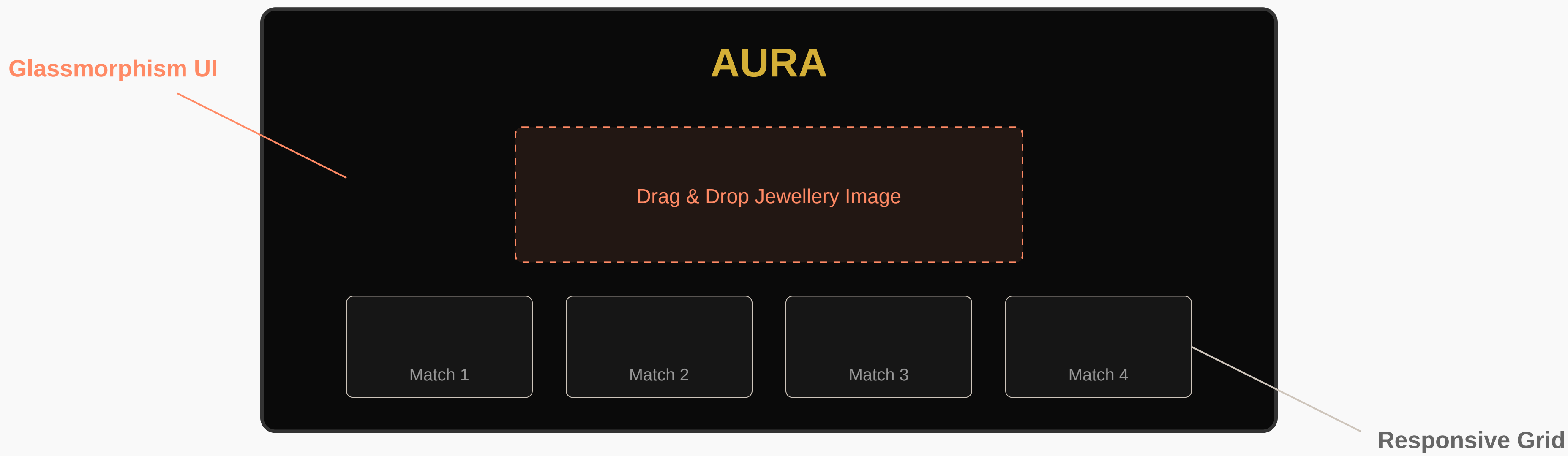
Frontend & Infra

USER EXPERIENCE

Vanilla JS (ES6+) for a lightweight glassmorphism UI and **Docker** for consistent containerized deployment.

Project Tech Stack: Python 3.10+, TensorFlow 2.x, FAISS-CPU, FastAPI, Docker

User Interface — Premium Experience



Primary Action (Upload) | Interactive Elements | Background / Secondary

Dark Luxury Aesthetic

Obsidian background (#0a0a0a) paired with champagne gold accents (#d4af37) and glassmorphism cards for a high-end jewellery feel.

Core UI Features

Drag-and-drop upload zone, instant local image preview via FileReader API, and animated result cards with similarity scores.

Seamless User Flow

Zero-friction experience: Upload -> Instant Preview -> Async Search -> Dynamic Results. Built with Vanilla JS for maximum performance.

Project Structure & Pipeline

Directory organization and 4-step automated execution timeline

```
AURA_PROJECT/
data/RingFIR/ ← 2605 images
frontend/
    index.html ← UI Markup
    style.css ← Glassmorphism
    app.js ← Search Logic
generate_embeddings.py
build_index.py
search.py
main.py ← FastAPI Server
features.npy ← 2605×1280 Matrix
faiss_index.bin ← Vector Index
requirements.txt
Dockerfile
```

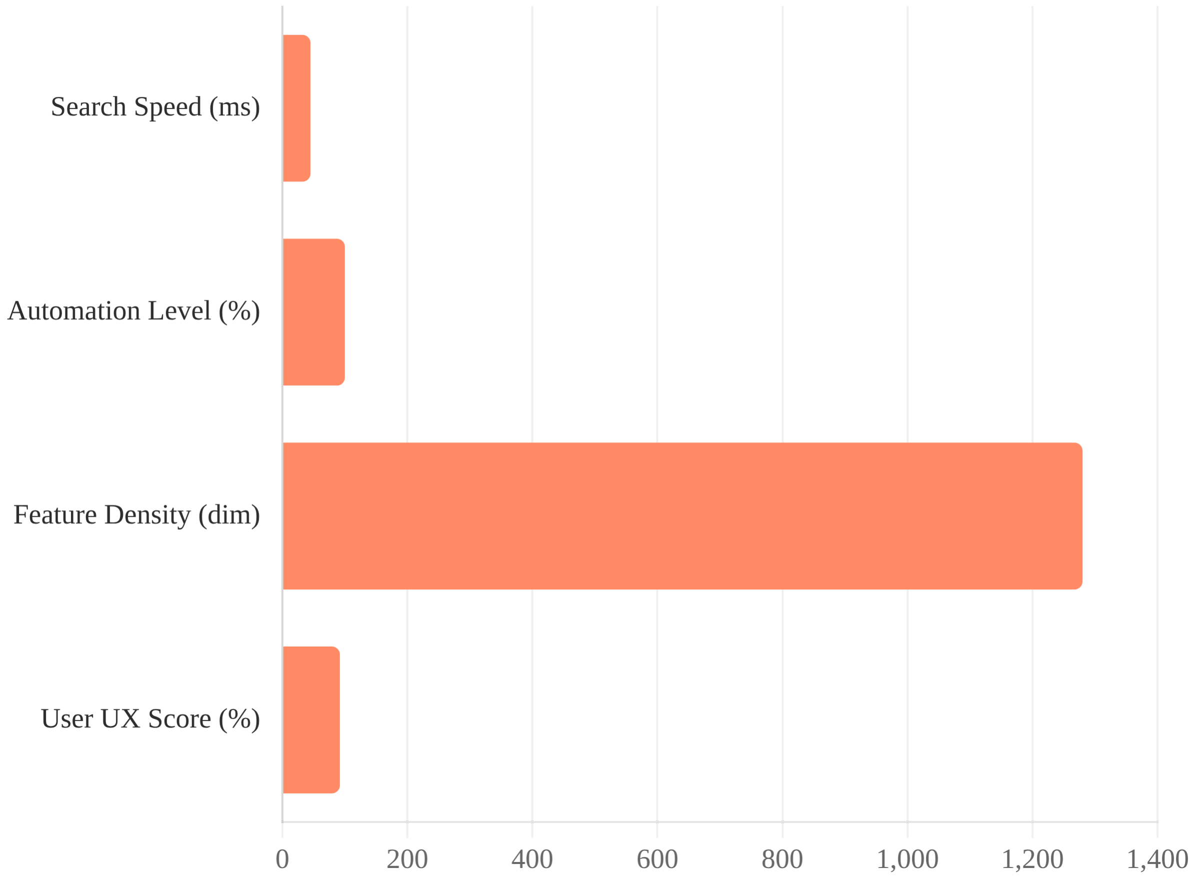
Step 1
Environment Setup
`pip install -r requirements.txt`

Step 2
Feature Extraction
`python generate_embeddings.py`

Step 3
FAISS Indexing
`python build_index.py`

Step 4
Live Deployment
`python main.py (FastAPI)`

Results & Conclusion



Source: AURA Internal Performance Benchmarking (v1.0)

Key Achievements

- **Real-time Search**: Achieved sub-50ms query response time for 2,605 images.
- **Zero Manual Tagging**: Fully automated pipeline from image to searchable vector.
- **High Precision**: EfficientNetB0 successfully captures intricate jewellery patterns.

Future Roadmap

- **Category Expansion**: Extend support to rings, necklaces, and bracelets.
- **Scaling**: Implement FAISS HNSW for indexing 10M+ items.
- **Multi-modal**: Combine visual search with natural language filters.

Thank You

AURA: Search with your eyes, not your words.

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AI-Powered Jewellery Visual Search Engine | Final Year Project | 2026